Transit Systems in Large Cities

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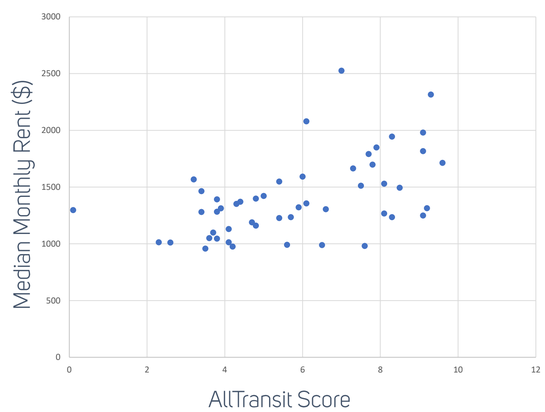
What might the quality of the transit systems in your city be telling you about the people who live there?

With rising economic stress in metropolitan areas, many are looking to public resources for relief. One such resources that plays a huge role in metropolitan areas is public transit. To begin to understand whether transit is affecting the quality of citizens day to day life we can test the correlation between quality of life and transit quality. Correlation is a measure between -1 and 1 which measures the direction and strength of linear relations between two variables where values near -1 and 1 represent strong relations and values near 0 represent weak relations.

Where is the data from?

The data used in this project is sourced from several organizations. Primarily, we are using data from the AllTransit project which was created and is maintained by The Center for Neighborhood Technology. CNT is a nationally recognized 501(c)(3) nonprofit based in Chicago focused on creating more livable metropolitan areas. The remainder of our data is sourced from a variety of government bodies including The Bureau of Labor Statistics for unemployment rates, The Center for Disease Control and Prevention for health data, The Federal Bureau of Investigation for crime data, and The Census Bureau for demographic data. It is worth noting here that the metropolitan statistical areas analyzed by each of these sources can vary slightly in their definitions.

AllTransit Score Against Median Monthly Rent



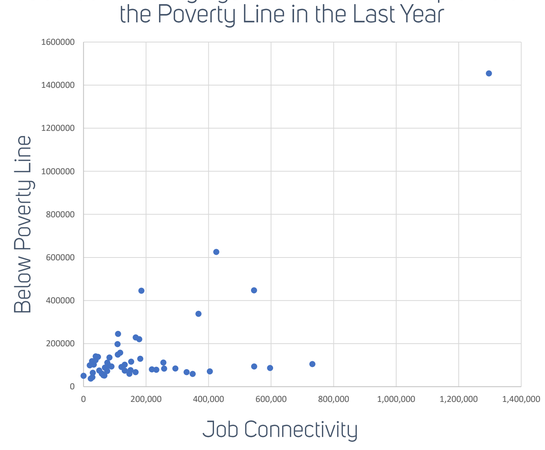
High Correlations

Our data yielded the highest correlations for AllTransit scores when compared to economic and miscellaneous indicators. These six indicators had a maximum correlation of 0.52 a median of 0.36 and a minimum of 0.17.

Above we can see the strongest correlation which is between AllTransit scores and median monthly rent. There is a clear upward trend in the data however it tends to become more loosely packed as rent and transit scores increase.

Further investigation reveals that the AllTransit metric which indicates how well transit systems connects people to their jobs yields strong correlations to the population under the poverty line, high school diploma attainment, and crime rate with the respective correlations being 0.71, 0.69, and 0.67.

Job Connectivity against Number of People Below the Poverty Line in the Last Year



The Indicators

In lieu of a single statistic which we could call quality of life we have chosen a several indicator statistics that can provide a more nuanced view of how transit relates to a citizens everyday life. These indicator statistics are organized into three categories as stated below.

For public health indicators, we are using lack of access to health insurance, lack of leisure time and physical activity, poor mental health, and the population percentage who've received their annual checkups.

For economic indicators, we are using unemployment rates, median monthly rent prices, median income, and the population below the poverty line.

Lastly, we are considering two miscellaneous indicators which might be seen as representing education and safety, respectively. These indicators are high school diploma attainment and crime rates.

Sources

Center for Neighborhood Technology. (2024). *Rankings*. AllTransit. https://alltransit.cnt.org/rankings/

Centers for Disease Control and Prevention. (2025, August 25). *500 cities: City-level data (GIS friendly format), 2019 release.* https://data.cdc.gov/500-Cities-Places/500-Cities-City-level-Data-GIS-Friendly-Format-201/dxpw-cm5u/about_data

U.S. Bureau of Labor Statistics. (2025, October 1). *Unemployment rates for metropolitan areas*. https://www.bls.gov/web/metro/laummtrk.htm

U.S. Census Bureau. (2025, July 1) *American Community Survey.* https://www2.census.gov

Federal Bureau of Investigation. (2025) *Uniform Crime Reporting Program.* https://cde.ucr.cjis.gov/LATEST/webapp/#/pages/downloads

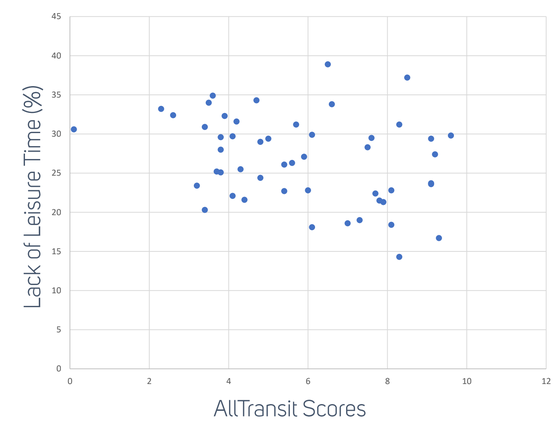
Conclusion

Overall, AllTransit scores did not have strong correlations to any individual indicators of quality of life we collected in our data. In the worst cases, the quality of transit had virtually no relationship to an indicator and in the best case, the quality of transit had a weak to moderate relationship to an indicator.

However, a more nuanced investigation of transit quality where certain aspects were centered almost unilaterally revealed stronger correlations. In particular, a transit systems size and ability to connect citizens to their jobs shows tends to yield higher correlations to our indicators.

This relation should be further investigated to determine the exact nature of the relationship between these aspects of transit and a citizens quality of life.

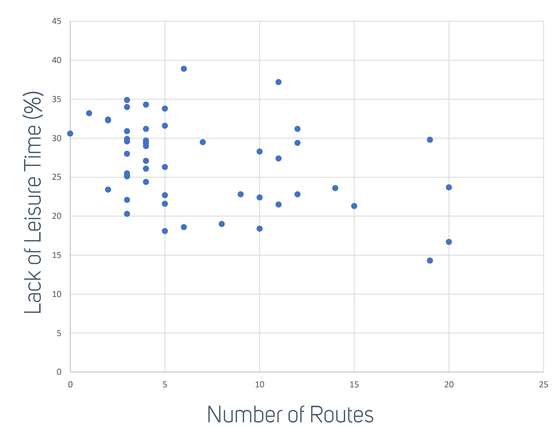
AllTransit Scores against Lack of Leisure Time



Low Correlations

Our data yielded lowest correlation to AllTranit scores among our health indicators. The maximum correlation being -0.3 the median being -0.25 and the minimum being -0.08.

Above, we can see the scatter plot for AllTransit scores and lack of leisure time. As indicated by its correlation value, it has a very slight negative trend and loosely packed data points which do not tend to cluster.

Further investigation reveals higher correlations between the number of routes in a transit system and our health indicators. The maximum correlation being -0.39 the median being -0.38 and the minimum being -0.05.

Number of Routes against Lack of Leisure Time